ASSIGNMENT 4

Relational Database vs. NoSQL Database

Online Shopping Management System

1. Introduction

Relational and NoSQL databases support different data-management needs. OnlineShoppingDB uses MySQL because customers, orders, order lines, products, and payments have structured relationships and require reliable integrity rules.

2. Comparison

Aspect	Relational database (MySQL)	NoSQL database (MongoDB)
Data model	Tables, rows, columns, defined schema, relationships and joins.	Document, key-value, wide-column, or graph models; often flexible schemas.
Scalability	Can scale vertically and horizontally with suitable architecture.	Many systems are designed for horizontal distribution; capabilities vary.
Consistency	MySQL/InnoDB supports ACID transactions and constraints.	Varies by database and configuration; some support transactions and configurable consistency.
Performance	Strong for structured queries, joins, and transactional workloads when designed well.	Can be efficient for access patterns aligned with its model; denormalized documents can reduce joins.
Schema changes	Managed through DDL and migrations.	Flexible document structures can simplify evolving records; validation is still useful.
Use cases	Order processing, banking, inventory, booking systems.	Flexible product metadata, content catalogs, event data, distributed workloads.

3. Why organizations choose different technologies

The choice depends on data shape, transaction needs, consistency, scale, latency, operational skills, cost, and reporting requirements. No single technology is best for every workload. Some organizations use more than one database for different application components.

4. Real-world examples and project fit

MySQL for this project: orders refer to customers; order lines refer to orders and products; payments refer to orders. Foreign keys, unique emails, checks, and transactions help protect relationships and support sales reports.

MongoDB-style document example: a product catalog may contain different attributes for phones, clothing, and furniture. A document model can accommodate varied attributes, although a relational design can also support them.

Banking and payment records are often suited to relational systems when consistent transactions and auditability are central. Large event or content workloads may use NoSQL when flexible data shapes and distribution are priorities. These are common patterns, not absolute rules.

5. Recommendation

Use MySQL for the core Online Shopping Management System because its data is structured and strongly connected. Consider NoSQL for a separate flexible product-catalog or clickstream workload only if measured requirements justify it.

6. Conclusion

Relational databases emphasize structured relationships and transactional integrity. NoSQL covers several data models suited to other data and scaling patterns. Organizations choose based on requirements, and hybrid architectures are possible.